

BI Insights Report

1. Executive Summary

Business context:

This analysis addresses the challenge of balancing product availability with waste reduction in a bakery environment.

Due to the perishable nature of bakery products, inaccurate demand estimation leads either to lost sales (underproduction) or increased waste (overproduction).

The dashboards were designed to monitor sales performance, demand stability, time-based consumption patterns, and waste-risk indicators to support data-driven daily decisions.

Dashboards used:

Monthly Product Demand Volatility, Product Revenue Contribution, Zero Sales Day Frequency, Sales by Time of Day, Total Revenue Weekdays vs Weekends .

Business value:

These dashboards transform raw sales data into actionable insights, allowing management to understand both product performance and customer demand patterns.

By linking metrics like demand volatility, zero-sales frequency, and revenue contribution to operational decisions, the bakery can optimize production, schedule staff effectively, and plan targeted promotions.

This proactive approach reduces waste, ensures product availability, and maximizes revenue throughout the year.

2. Dashboard-Based Key Findings

2.1 Product Performance & Demand Stability

Related dashboard / visual(s):

Monthly Product Demand Volatility (area chart)

What the dashboard shows (facts): The chart highlights how demand evolves throughout the year and it can be divided into periods by demand level:

From January to April = Steady Growth: Demand rises from roughly 2.1K units to nearly 3K, suggesting a strong spring season.

From April to June = Mid-Year Stability: Demand stabilizes around the 2.9K mark during early summer.

July and August = Peak Season : The highest demand occurs in August (Month 8), reaching approximately 3.6K units.

From September to December = Low-Demand period: There is a significant and rapid drop-off starting in September. By October, demand crashes to roughly 1.4K units, ending the year at its lowest point (1.1K).

Interpretation (insight):

The pattern indicates high seasonality as product demand is not stable across the year.

It is strong in mid-year and substantially weaker in year-end.

KPI referenced:

Product Demand Volatility

Business implication:

Production planning should be dynamically adjusted: higher output during mid-year peak months and reduced batch sizes in the final quarter to avoid overproduction.

From October to December (low-demand months), the bakery should adopt conservative production strategies, limited assortments, or demand-stimulating actions (example: promotions) to mitigate revenue loss and waste.

2.2 Product Revenue Contribution

Related dashboard / visual(s):

Product Revenue Contribution (donut chart)

What the dashboard shows (facts):

The Traditional Baguette is the dominant product, contributing 42.61% of total revenue on its own.

The next highest contributors are Formule Sandwich (10.37%), Croissant (9.89%), and Pain au Chocolat (9.15%).

Together, the top four products generate over 70% of total revenue, indicating strong revenue concentration.

The remaining products (Banette, Baguette, Sandwich Complet, Special Bread, Traiteur, Grand Far Breton) each contribute less than 8% individually, forming a long tail of lower-revenue items.

Interpretation (insight):

Revenue is highly concentrated in a small number of core products, with the Traditional Baguette acting as the primary revenue driver. While product variety is wide, most items contribute marginally compared to the leading products.

→ This suggests that customer demand is focused and predictable, rather than evenly distributed across the assortment.

KPI(s) referenced:

Daily Product Sales, Total Units Sold

Business implication:

Core products, especially the Traditional Baguette, must be prioritized in production planning, as any stockout would have a disproportionate impact on revenue.

Low-contribution products increase assortment complexity and may elevate waste risk if produced at similar levels to top sellers.

The bakery should consider rationalizing or limiting production volumes of long-tail products, particularly during low-demand periods, while protecting availability of top contributors.

2.3 Time-of-Day Sales Behavior

Related dashboard / visual(s):

Sales by Time of Day (Bar Chart)

What the dashboard shows (facts):

- Morning sales dominate, generating by far the highest share of total revenue.
- Afternoon sales drop significantly compared to the morning but still contribute a meaningful portion.
- Evening sales are minimal, representing only a small fraction of daily revenue.

Interpretation (insight):

→ Most purchases occur in the morning (during the day).

KPI(s) referenced:

Sales by Time of Day, Late-Day Sales Ratio

Business implication:

The bakery should prioritize fresh production in the morning when demand is highest.

Since evening sales are very low, products produced too late in the day have a high probability of remaining unsold, increasing waste.

Afternoon and evening production should be scaled down or limited to fast-moving products only.

2.4 Weekly & Seasonal Patterns

Related dashboard / visual(s):

Sales by Weekday (pie chart)

What the dashboard shows (facts):

The pie chart shows that weekdays account for 60.66% of total revenue, while weekends contribute 39.34%.

Interpretation (insight):

Demand is therefore more evenly distributed across regular working days, rather than being concentrated on weekends.

KPI(s) referenced:

Total Sales Revenue

Business implication:

Since weekdays generate most of the revenue, the bakery must be consistent throughout the weekdays not only the weekends.

Targeted promotions and special product offerings (example: Weekend coffee+pastry combo: a special price for: 1 coffee + 1 pastry (weekend-only pricing)).

2.5 Waste Risk Indicators

Related dashboard / visual(s):

Zero-Sale Days Table

What the dashboard shows (facts):

The dashboard lists various food products and assigns them a value for Zero Sales Day Frequency. This metric represents the proportion of days on which a specific item had zero sales.

Scale: The values range from 0.00 to 1.00.

High Frequency (1.00): Items like *Cake*, *Pain Noir*, and *Tartelette Cocktail* have a frequency of 1.00, meaning they are almost never sold on a daily basis (or are only available for special orders).

Low Frequency (0.00): Items like *Croissant*, *Pain au Chocolat*, and *Special Bread* have a frequency of 0.00, meaning they are sold every single day without exception.

Interpretation (insight):

The data categorizes products into three distinct performance tiers:

1. **Core Daily Drivers (0.00 – 0.10):** Products like Baguette, Pain, and Croissant. These are the "staples" that bring customers into the shop daily.
2. **Moderate/Occasional Sellers (0.11 – 0.80):** Items like Eclair (0.22), Flan (0.33), and Mille Feuilles (0.47). These products sell often but are not guaranteed to move every day.
3. **Slow Movers / Niche Items (0.81 – 1.00):** A large volume of specialty items such as Galette 8 Pers, Reduction Sucrees, and various Plat Prepare. These items have a very high "zero-day" rate, suggesting they are either highly seasonal, expensive, or made-to-order.

KPI(s) referenced:

Zero-Sale Day Frequency

Business implication:

Items with high frequency (0.90+) that are fresh-baked present a massive risk for food waste (shrinkage). The business should consider moving these to a "pre-order only" model to protect margins.

The ~0.00 items are the engine of the business. Stockouts of these items (like Croissants) would likely result in immediate lost revenue and customer dissatisfaction, as these are the primary reasons for foot traffic.

The dashboard reveals a long tail of products that rarely sell. The business could likely simplify its operations and reduce labor costs by cutting the bottom 20% of items that have a frequency of 0.95 or higher.

For items in the middle range (example: Brioche at 0.15), the bakery can use this data to fine-tune production volumes, baking smaller batches to ensure they sell out rather than risking a zero sale day of leftover stock.

3. Why These Results Matter

Dashboards reduce uncertainty by transforming historical sales into visual demand signals.

KPIs such as zero-sale days allow early detection of waste risk before losses occur.

Visual monitoring is essential for perishable goods, where delayed action directly translates into financial loss.

4. Business Recommendations (Dashboard-Driven)

Recommendation 1

Action: Introduce phased baking schedules.

Supporting dashboard(s): Time-of-Day Sales

KPIs impacted: Late-Day Sales Ratio

Expected benefit: Improved product freshness and reduced end-of-day leftovers.

Recommendation 2

Action: Focus assortment on high-contribution products.

Supporting dashboard(s): Product Revenue Contribution

KPIs impacted: Revenue per Product

Expected benefit: Higher profitability with simpler operations.

Recommendation 3

Action: Move high zero-sales-frequency items (0.90+) to a pre-order or made-to-order model.

Supporting dashboard(s): Zero Sales Day Frequency table

KPIs impacted: Zero Sales Day frequency

Expected benefit: Reduced food waste, lower production costs, and higher profit margins on specialty items.

Recommendation 4

Action: Adjust production volumes dynamically according to seasonal demand patterns.

Supporting dashboard(s): Monthly Product Demand Volatility (area chart)

KPIs impacted: Product Demand Volatility

Expected benefit: Reduced overproduction during low-demand months, optimized use of resources, and minimized waste.

Recommendation 5

Action: Introduce targeted weekend promotion: coffee + pastry combos at a special price.

Supporting dashboard(s): Sales by Weekday (pie chart)

KPIs impacted: Unit sales

Expected benefit: Increase weekend sales and attract more customers during slower periods relative to weekdays.

5. Limitations (Data & Dashboard Scope)

- Waste is inferred through sales behavior, not directly measured.
 - Production quantities and disposal data are not captured.
 - Promotional effects are not explicitly modeled.
 - Analysis is limited to the available time granularity.
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6. Future Improvements

- Add a direct waste tracking dashboard.
 - Integrate production and inventory data.
 - Introduce demand forecasting models.
 - Apply optimization techniques for batch sizing.
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7. Conclusion

The Power BI dashboards successfully transform transactional bakery data into operational intelligence.

By visually linking demand patterns, timing effects, and waste-risk indicators, the analysis enables actionable, data-driven decisions that directly support profitability and waste reduction.